

Recent advances in lattice thermal conductivity calculation using machine-learning interatomic potentials ^{EP}

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ABSTRACT

The accuracy of the interatomic potential functions employed in molecular dynamics (MD) simulation is one of the most important challenges of this technique. In contrast, the high accuracy *ab initio* quantum simulation cannot be an alternative to MD due to its high computational cost. In the meantime, the machine learning approach has been able to compromise these two numerical techniques. This work unveils how the MD interatomic potentials have been improved through training over *ab initio* datasets and are able to well calculate phononic thermal transport of materials. Therefore, this powerful tool allows the quantum computational order accuracy with a timescale in the order of classical computations. Besides, the thermal conductivity of a few 2D and 3D structures, which have been calculated using machine learning interatomic potentials (MLIPs), is presented and compared with experimental and quantum counterparts. Finally, it is discussed that how MLIPs can be developed not only to estimate other properties of pristine materials, such as mechanical properties, but also to predict the properties of defective materials.

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I. INTRODUCTION

Heat transfer in two-dimensional (2D) and three-dimensional (3D) materials due to their application in nanoelectronics and batteries is a determining factor in their design and optimal performance.^{1–3} With the development of experimental methods, the number of 2D and 3D materials synthesized in the laboratory has increased.^{4–6} In addition, theoretical methods have predicted a large number of stable 2D materials.^{7–9} Therefore, having fast and reliable methods to know the thermal properties of 2D and 3D materials is very important.

There are both experimental and theoretical methods for calculating thermal properties, especially phononic thermal conductivity, which is the most important thermal property in 2D semiconductor materials. Among the theoretical methods, numerical techniques attracted the attention of researchers due to their high flexibility in calculating the thermal conductivity of a wide range of materials. In general, popular numerical methods in calculating thermal conductivity are divided into two main branches:

Molecular Dynamics (MD) simulation and the Boltzmann Transport Equation (BTE). However, there are other methods such as Monte-Carlo simulation¹⁰ that are less considered due to their limitations. In the MD approach, thermal conductivity is calculated by selecting an appropriate potential function to describe interatomic forces. In the BTE method, thermal conductivity is calculated by having interatomic forces usually obtained from quantum calculations.

The main limitation of the MD simulation method for thermal conductivity is the insufficient accuracy of the potential function, which is usually sacrificed for the simplicity of the potential form and the computational cost. Also, the heavy calculations of quantum computing to obtain the phononic thermal properties, even using the Density Function Theory (DFT) approximation, are among the limitations of the quantum methods utilized in the BTE method. Recently, with the advent of artificial intelligence and machine learning, it has become possible to significantly improve the accuracy of potential functions used in MD simulations in

solving Newton's equations of motion and in the calculation of interatomic force constants required in the BTE method. Machine-learning interatomic potentials (MLIPs) offer a unique possibility to conduct first-principles multiscale modeling, in which *ab initio* level of accuracy can be hierarchically bridged to explore lattice thermal conductivity. The main steps within the first-principles hierarchical multiscale modeling framework include four key steps: (1) DFT simulations, (2) development of MLIPs, (3) classical MD simulations, and (4) calculation of effective lattice thermal conductivity. In some earlier studies,^{11,12} it is shown that MLIPs can be used to conduct first-principles multiscale modeling of lattice thermal conductivity or the mechanical/failure response of macroscopic systems.

In addition to developing the potential functions, machine learning methods can also be implemented directly in predicting the properties of the material.^{15–17} Chen *et al.*¹⁸ provided a machine learning-based model using the Gaussian process regression algorithm for predicting the lattice thermal conductivity of inorganic materials. The benchmark dataset in this study is the thermal conductivity of 100 samples computed experimentally. In another study, based on the *K*-nearest neighbor, fixed-radius nearest neighbor, and Gaussian processes machine learning regression methods, Goodarzi and Bahramian¹⁹ developed models for determining the thermal conductivity coefficient of polymeric aerogels. Estimating the coefficients of the convection heat transfer using a random forest machine learning algorithm was also presented by Kwon *et al.*²⁰

In this article, we intend to briefly introduce the potential functions utilized in the machine learning method and examine how they are trained by *ab initio* molecular dynamics (AIMD). Then, we present how to compute the lattice thermal conductivity in MD simulation and in BTE method and review the recent results in obtaining the thermal conductivity of various materials using machine learning interatomic potentials. Comparing their accuracy with DFT-BTE and experimental counterparts and the advantages and disadvantages of using different potential formats will also be reviewed and discussed.

II. MACHINE-LEARNING INTERATOMIC POTENTIAL

In recent years, machine learning has been extensively used to accelerate the recognition of the physical properties of various materials.^{21–24} A simple case is the application of machine learning-based potential to molecular dynamics simulations by using more accurate interatomic forces and energies than that from the empirical potentials.^{25–28}

MLIPs are a type of nonparametric designed interatomic potentials with the aim of providing accuracy in the order of quantum mechanics computations, while their computational costs are in the order of empirical interatomic potentials. Based on the terminology of the regression analysis, all potentials can be classified into two categories, parametric and nonparametric potentials.²⁹ The main difference between these two types of potentials is the ability of nonparametric potentials to improve their accuracy systematically. In order to achieve high accuracy in rather large systems, nonparametric potentials are more suitable choices. However, it should be noted that reaching such a high accuracy

requires higher computational costs. Furthermore, the performance of these potentials is restricted to the reference quantum mechanics model to which the potential is fitted. In other words, each fitted potential cannot be used for every situation.²⁹ A nonparametric potential consists of two basic elements. The first element is called “descriptors” and the second element is a regression model that is a function of the descriptors. The goal of every machine-learning potential is the determination of the potential energy surface (PES) that is related to descriptors. The descriptors are demonstrative of the atomic environment and basically have to be invariant, and in a system with different types of atoms, they should be able to recognize them. Usually, a cut-off function is considered in descriptors to guarantee that the descriptors work within a specific cut-off radius. Behler-Parrinello descriptors³⁰ and bispectrum coefficients³¹ as the atom-centered descriptors are among the most popular descriptors in developing the MLIPs.³² The selection of descriptors has a major influence on the performance of machine-learning potential.³¹ The regression model as another basic element in nonparametric potentials is a methodology used to map the descriptors to the PES.³¹ There are various regression methods among which a high-dimensional neural network,^{30,33} Gaussian process regression,^{34,35} and linear/polynomial regression^{29,31,36} have been well implemented in MLIPs.³² Spectral Neighbor Analysis Potential (SNAP),³¹ Moment Tensor Potentials (MTPs),²⁹ Neural Network Potentials (NNP),³⁷ and Gaussian Approximation Potentials (GAP)³⁴ are four conventional approaches in creating nonparametric potentials. NNP and GAP, respectively, use artificial neural networks and Gaussian process regression models, and Behler-Parrinello and smooth-overlap of atomic positions (SOAP) kernel descriptors. In the SNAP, the descriptors are the bispectrum coefficients of the neighbor density and the relationship between the energy/forces with the bispectrum components is considered linearly. A modified version of SNAP with quadratic relationships has been also proposed in which a relatively small increase in computational cost leads to a considerable increase in accuracy.³⁶ MTP is based on linear regression and invariant polynomials, and the moment tensor descriptors are used to define the atomic environments.

In order to facilitate the use of these machine learning methods, several open-source software packages have been developed in recent years. These packages were written in different programming languages such as Python (PyXtal_FF,³⁸ QUIP, SIMPLE-NN,³⁹ and AMP⁴⁰), C/C++ (ænet,⁴¹ n2p2,⁴² and SIMPLE-NN³⁹), and FORTRAN (RubNNet4MD,⁴³ RuNNer,^{44,45} ænet,⁴¹ and AMP⁴⁰), and some of them include various descriptors and regression models that enable them to be used for a wide range of potential training.^{38,40} Table I provides a summary of the characteristics of four popular machine learning methods in creating interatomic potentials and their prepared software packages. It is worth mentioning that the investigation of the combination of various descriptors and regression models in addition to the mentioned common methods has also been proposed.^{38,46}

III. POTENTIAL FUNCTION TRAINING

This section discusses how a potential function is trained by machine learning to predict the thermal conductivity of 2D and 3D

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TABLE I. Machine learning methods in creating interatomic potentials.

Method	Descriptor	Regression model	Package
MTP ²⁹	Moment tensor	Linear	MLIP ²²
SNAP ³¹	Bispectrum coefficients	Linear	PyXtal_FF ³⁸
GAP ³⁴	SOAP kernel	Gaussian process	QUIP ³⁵
NNP ³⁷	Behler–Parrinello	Artificial neural networks	RuNNer ^{44,45} RubNNet4MD ⁴³ ænet ⁴¹ n2p2 ⁴² SIMPLE-NN ³⁹ PyXtal_FF ³⁸

materials based on experimental or theoretical calculation data. In order to train the potential function, in the first step, the structure is optimized by *ab initio* quantum calculations to eliminate unnecessary stresses and place the atoms in a position where the structure is at the lowest energy level. Then, depending on the type of study, the structure at different temperatures (to calculate the thermal conductivity) or strains (to calculate the elastic modulus) or different defects (to study the defective structure) is simulated using AIMD in time interval of about 1 ps usually.^{47–52} Because the results obtained by AIMD are interconnected at small times, a percentage of the results (usually 10%) are regularly utilized as a sub-sample to train the desired potential (usually using known packages such as MLIP^{22,29} and deepMD⁵³ or other packages mentioned in Table I). For training, potential coefficients are determined in such a way that the difference between the predicted value and the value obtained from the first-principles calculations for energy, forces, and stresses is minimized.^{22,29}

$$\sum_{k=1}^K \left[w_e (E_k^{\text{AIMD}} - E_k^{\text{Pot}})^2 + w_f \sum_i^N |f_{k,i}^{\text{AIMD}} - f_{k,i}^{\text{Pot}}|^2 + w_s \sum_{i,j=1}^3 |\sigma_{k,ij}^{\text{AIMD}} - \sigma_{k,ij}^{\text{Pot}}|^2 \right] \rightarrow \min, \quad (1)$$

where E_k^{AIMD} , $f_{k,i}^{\text{AIMD}}$, and $\sigma_{k,ij}^{\text{AIMD}}$ represent the energy, atomic forces, and stresses, respectively; E_k^{Pot} , $f_{k,i}^{\text{Pot}}$, and $\sigma_{k,ij}^{\text{Pot}}$ denote the identical values computed by MLIP; K is the number of configurations; that is, the number of first-principles calculations in creating the training set. Also, N corresponds to the number of atoms, and w_e , w_f , and w_s refer to the positive weights that demonstrate the significance of energies, forces, and stresses in optimization, respectively. The variety and size of configurations in the training set are the contributing factors in the selection of weighting coefficients. So, for the training set including configurations with no more than tens of atoms and those whose unit cell and stresses do not have a dominant effect, the coefficient of w_s is considered zero. For a training set in which configurations have different sizes such as the same structures with different supercells, the influence of size can be eliminated by considering the number of atoms as the scaling factor in weighting coefficients.²²

The trained potential is then retested by all AIMD results, and points that are significantly outside the potential range are added to

the new percentage of subsets. Eventually, the potential is re-trained based on the new sub-paradigm. Finally, this potential can be used to continue working. Figure 1 summarizes the potential function training flowchart.^{46,54} Also, some references associated with each trajectory are presented in Table II. Some freely accessible datasets on training the potential functions and using them in MD simulations were provided by Mortazavi *et al.*^{55,56}

IV. THERMAL CONDUCTIVITY CALCULATION METHODS

Equilibrium molecular dynamics (EMD) and non-equilibrium molecular dynamics (NEMD) methods can be employed to calculate thermal conductivity. “Direct method”^{61–63} and “Green–Kubo”^{63,64} are the two most common methods to compute the thermal conductivity. To utilize the NEMD method, a temperature gradient is applied to the simulation cell, which is a resembling mechanism of measurement in experimental methods. The Green–Kubo method is an EMD simulation that can be used by employing the fluctuation-dissipation theorem.⁶⁵ Another well-known numerical technique to study the thermal transport is the BTE method that is based on interatomic force calculations.

A. Non-equilibrium molecular dynamics

There are multiple methods to calculate thermal conductivity by NEMD. In the first method, two regions are created and their desired temperatures are preserved by adding and subtracting energy to the hot zone and from the cold zone, respectively.^{66,67} The other method is to add and remove a known amount of energy to and from two different regions.⁶⁸ Afterward, the temperature gradient in the material is calculated to estimate the thermal conductivity. A reverse non-equilibrium molecular dynamics (RNEMD) as another method is performed using an algorithm suggested by Müller-Plathe.⁶⁹ In this method, kinetic energy displacement is imposed to create a temperature gradient within the specimen.

The thermal conductivity (k) is inferred based on the Fourier’s law as follows:

$$J = -k \nabla T, \quad (2)$$

where J is the heat flux, and ∇T is the temperature gradient.

B. Equilibrium molecular dynamics

In the EMD method that is based on the linear response theory, a system is maintained at the equilibrium state. In the linear response theory, thermal conductivity can be extracted based on the Green–Kubo formalism⁷⁰ as follows:

$$k = \frac{V}{3k_B T^2} \int_0^\infty \langle J(t)J(0) \rangle dt, \quad (3)$$

where k represents thermal conductivity, V denotes the simulation box volume, T refers to the system temperature, k_B indicates Boltzmann’s constant, J denotes the heat flux, and $\langle J(t)J(0) \rangle$ is heat-current autocorrelation function (HCACF).

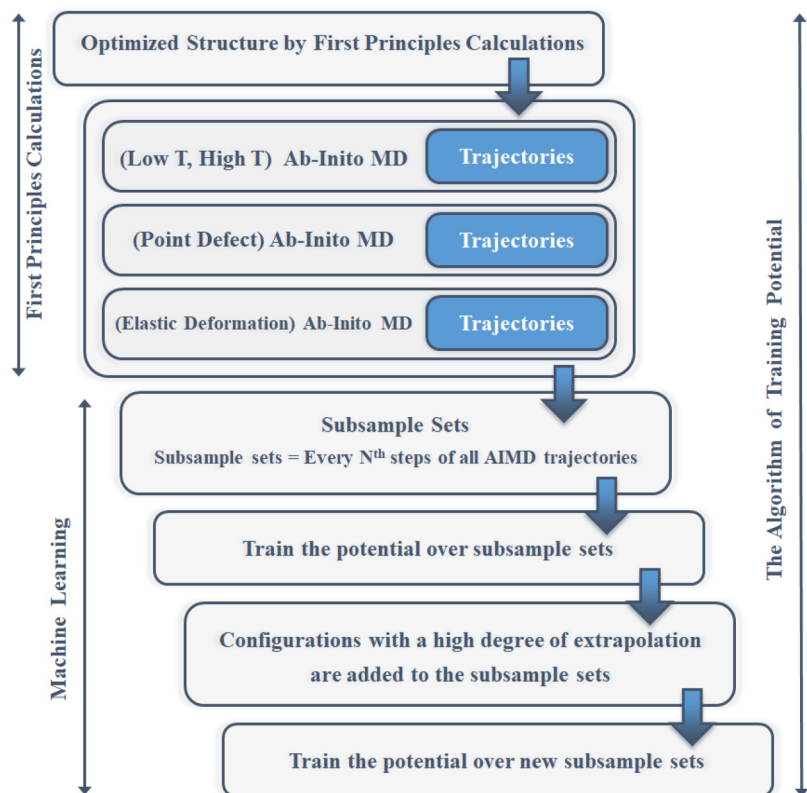


FIG. 1. The algorithm of training potential function using AIMD trajectories.

Within the length of integral time, HCACF must deteriorate to zero to produce data convergence. Therefore, the correlation length is an important parameter utilized to guarantee that there is enough time for the autocorrelation function to deteriorate to zero.^{71,72}

C. Boltzmann transport equation

In materials with thermal gradient ∇T inside them, heat transport commonly occurs by two thermal barriers that are electrons and crystal lattice vibrations. In the case of semiconductors, the transported heat by the electrons is smaller than that of lattice vibrations, which are quantized into phonons, and the dominant thermal transport mechanism. According to the assumption of BTE, phonon occupation changes in a small volume of the crystal with a temperature gradient across it due to two main reasons:

TABLE II. Some research studies associated with the mentioned trajectories in Fig. 1.

Trajectories	Related research
Low T, High T	Arabha <i>et al.</i> , ⁵⁴ Mortazavi <i>et al.</i> , ^{57–59} Korotaev <i>et al.</i> , ⁴⁸ Ladygin <i>et al.</i> , ⁴⁹ Dragoni <i>et al.</i> ⁶⁰
Point defects	Arabha <i>et al.</i> , ⁵⁴ Mortazavi <i>et al.</i> , ¹² Dragoni <i>et al.</i> ⁶⁰
Elastic Deformation	Arabha <i>et al.</i> , ⁵⁴ Mortazavi <i>et al.</i> , ¹² Dragoni <i>et al.</i> ⁶⁰

phonon drift and collisions. Different types of collisions might happen, such as phonons scattering by other phonons, crystal boundaries, defects, and impurities. No alteration in the number of phonons in the small volume of a crystal is expected in the case of a steady state, and therefore, the total amount of drift and collision terms must be zero,

$$\left. \frac{\partial n(\vec{q})}{\partial t} \right|_c + \vec{v}(\vec{q}) \cdot \nabla T \frac{\partial n(\vec{q})}{\partial T} = 0, \quad (4)$$

where n is the phonon distribution function, $\vec{q}$ is the phonon wave vector, and $\vec{v}$ is the group velocity. $\nabla n(\vec{q}) \cdot \vec{v}(\vec{q})$ corresponds to the distribution function change rate caused by phonon motions, and $[\partial n(\vec{q})/\partial t]_c$ refers to the same variation rate due to collisions, leading to a steady-state condition.⁷³

Different packages nowadays give the ability to estimate thermal conductivity using the BTE method. For instance, the ShengBTE package⁷⁴ provides the ability to determine the lattice thermal conductivity by solving the BTE utilizing a set of interatomic force constants (IFCs). ShengBTE requires both harmonic (second order) and anharmonic (third order) IFCs to determine lattice thermal conductivity. To obtain anharmonic IFCs, depending on the symmetry and cutoff distance, a few hundred to several thousand force calculations over relatively large supercell structures have to be performed, which can be computationally exceedingly

expensive with conventional DFT methods, using the Quantum Espresso⁷⁵ and VASP.^{76,77} MLIPs can accelerate this process and save a great deal of time and computational cost by substituting DFT simulations, as have been proven in numerous recent studies.

Recently, it has been shown that considering four-phonon scattering has a significant contribution to the thermal conductivity of some structures. Calculating this phonon scattering by DFT calculations is very computationally demanding. It has been shown that MLIP could calculate this phonon scattering with low computational cost.^{57,78}

V. THERMAL CONDUCTIVITY OF MATERIALS

In this section, we review the thermal conductivity of 2D and 3D materials and compare the results of experiments, BTE–DFT, and classical MD with the corresponding values that are reported using the MLIPs approach. Table III shows the thermal conductivity of 2D and 3D materials at room temperature by experiment, BTE–DFT, classical MD, and MLIP approaches.

A. Thermal conductivity of 2D materials

1. Graphene

Since the emergence of graphene,¹¹⁹ several experimental and numerical studies have been performed to study its thermal properties. The experimental measurements of thermal conductivities of single-layer graphene are mostly estimated within 1500–5300 W/m K.^{79–85} Figure 2 compares the thermal conductivity of graphene by MLIPs, experiment, BTE–DFT, and classical MD approach.

The existence of multiple uncertainty sources in the thermal conductivity calculation of 2D materials usually causes scattering of reported values. As an alternative, numerical modeling approaches, i.e., DFT and MD, have been developed for the accurate estimation of the thermal conductivity of 2D materials. Since the accuracy of the MD results mainly depend on the interatomic potentials, the DFT and BTE combination is a more reliable approach to calculate the 2D and 3D materials’ thermal conductivity. Numerous DFT calculations have been shown a range of 1934–5500 W/m K for the thermal conductivity of graphene (Fig. 2 and Table III).

TABLE III. Thermal conductivity of 2D and 3D materials at room temperature by experiment, BTE–DFT, classical MD, and MLIP approaches. The unit is W/m K.

Materials	Machine learning		Other computational methods		Experiment
	Value	Method	Classical MD	BTE–DFT	
2D Materials					
Graphene	3600 ¹¹	BTE	1600 ⁹¹ 3000 ± 100 ⁹²	3720 ⁸⁶ 3600 ⁸⁷ 3150 ⁸⁸ 1936 ⁸⁹ 5500 ⁹⁰ 3288 ⁹⁰	~3080–5300 ⁷⁹ 4127 ± 539 ⁸⁰ 1500–5000 ⁸¹ 2500 ⁸² 2430 ± 190 ⁸³ ~1800 ⁸⁴ 1937–2298 ⁸⁵
Borophene (Zigzag)	52 ¹¹	NEMD	45.3 ± 5.2 ⁶⁶ , 4.2 ⁹⁴	~79.3 (W/K) ⁹³	...
Borophene (Armchair)	112 ¹¹		85.2 ± 2.4 ⁶⁶ 83.5 ⁹⁴	~144.8 (W/K) ⁹³	...
C ₂ N	85.5 ± 5 ⁵⁴	NEMD	63.3 ± 5 ⁵⁴ ~67.38 ⁶⁷	82.2 ⁹⁵	...
C ₃ N	418 ± 20 ⁵⁹	NEMD	~779.5 ⁶⁷ 810 ⁹⁷	380 ⁹⁶	...
MoS ₂	~155 ⁵⁷ 89 ± 11 ¹⁰¹	BTE EMD	38.1 ¹⁰⁰	~67 ⁹⁸ ~130 ⁹⁹	...
Silicene	~26 ⁵⁷	BTE	41 ¹⁰⁴ ; 40 ¹⁰⁵	~27 ¹⁰² ~25 ¹⁰³	...
Phosphorene (Zigzag)	~19 ⁵⁷	BTE	33.0 ¹⁰⁸ ~63.6 ¹⁰⁹ 152.7 ¹⁰⁸ ~110.7 ¹⁰⁹	~21 ¹⁰⁶ ~24 ¹⁰⁷	...
Phosphorene (Armchair)	~87 ⁵⁷			~97 ¹⁰⁶ 112 ¹⁰⁷	...
3D Materials					
Diamond	~2500 ⁵⁷	BTE	...	~2400 ¹¹²	~2500 ¹¹⁰ ~2450 ¹¹¹
Silicon	~145 ⁵⁷ 121 ¹¹⁶	BTE EMD	~240 ¹¹⁵	127–148 ¹¹⁴	~157 ¹¹³
Bas	~2500 ⁵⁷	BTE	~2400 ¹¹⁷	~2250 ¹¹²	...
InAs	~30 ⁵⁷	BTE	...		~24 ¹¹⁸

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Because DFT-based simulations are too costly in terms of energy and time, a new approach has been proposed to employ MLIP as a substitution for the computationally demanding DFT calculations in evaluating the thermal conductivity of materials. For graphene, the thermal conductivity at room temperature is estimated to be 3600 W/m K^{11} using MLIPs (Fig. 2 and Table III). Besides, it has shown that MLIPs can also well predict the thermal conductivity of graphene at different temperatures and its mean free path as well (Fig. 3).⁵⁷

One of the factors that influence the thermal conductivity obtained via the BTE-DFT method is the correct choice of exchange-correlation functions. For instance, Taheri *et al.* showed that the thermal conductivity of graphene is varied by the changing exchange-correlation function.¹²¹ However, they have recently shown that choosing an exchange-correlation function does not affect thermal conductivity. However, Mortazavi *et al.* showed that in calculating thermal conductivity by the MLIP-BTE method, the selection of the exchange-correlation function, which appears in *ab initio* modeling in the MLIP training stage, has no effect on the thermal conductivity results.⁵⁷

2. Other 2D materials

The MLIPs have also been employed to study thermal conductivity and phononic properties of other 2D materials,⁵⁸ i.e., nitrogenated holey graphene (C_2N), molybdenum disulfide (MoS_2), borophene, phosphorene, silicene, and C_3N . As it can be seen in Fig. 4 and Table III, the MLIPs have been shown to produce thermal conductivity close to DFT results.

Carbon-based 2D materials, such as C_3N and C_2N , are among the most interesting materials because of their wide range of applications. The thermal conductivity of C_3N has been estimated as 380^{96} and $779.5\text{--}810 \text{ W/m K}^{67,97}$ using DFT and classical MD,

respectively. Using MLIPs, the thermal conductivity of C_3N was predicted to be $418 \pm 20 \text{ W/m K}^{59}$, which is 10% far from DFT results. For C_2N , this difference between DFT and MLIPs results is less than 5%. While the thermal conductivity of C_2N was predicted to be 63.3 ± 5^{54} and $67.4 \text{ W/m K}^{67,97}$ using classical MD, it was reported as 82.2^{95} and $85.5 \pm 5 \text{ W/m K}^{19}$ using DFT and MLIP approaches, respectively (Table III and Fig. 4).

Among 2D materials, there are materials with anisotropic thermal properties due to their anisotropic atomic structure, such as phosphorene and borophene. It has been shown that the MLIPs' approach could well estimate the thermal conductivity of these anisotropic 2D materials. Borophene has been shown to have $\sim 53\%$ anisotropy in the thermal conductivity (armchair: 112 W/m K and zigzag: 52 W/m K)¹¹ and phosphorene $\sim 78\%$ (armchair: 87 W/m K and zigzag: 19 W/m K)⁵⁷ using MLIPs. The thermal conductivity anisotropy of borophene was estimated as $\sim 45\%$ ⁹³ and $\sim 47\%$ ⁶⁶ and for phosphorene as $\sim 77\%$ ¹⁰⁷ and $\sim 78\%$ ¹⁰⁹ employing DFT and classical MD, respectively (Table III and Fig. 4).

A recent study of the lattice thermal conductivity of graphene/borophene heterostructures revealed that MLIPs trained over short AIMD trajectories enable efficient first-principles modeling of complex 2D structures.¹¹ The MLIP not only predicts the lattice thermal conductivity of pristine phases of borophene and graphene efficiently, it well estimates the interfacial thermal conductance of complex borophene/graphene (Fig. 5).¹¹

It is worth mentioning that to assess the thermal properties, DFT calculations are commonly executed using supercell structures. DFT is a computationally efficient approach for the majority of highly symmetrical lattices. By employing modifications like increasing the supercell size, increasing plane-wave cutoff energy, or improving the resolution of k-point grids, DFT calculations can offer a suitable accuracy for highly symmetrical lattices like graphene, but for less symmetrical and nanoporous lattices may

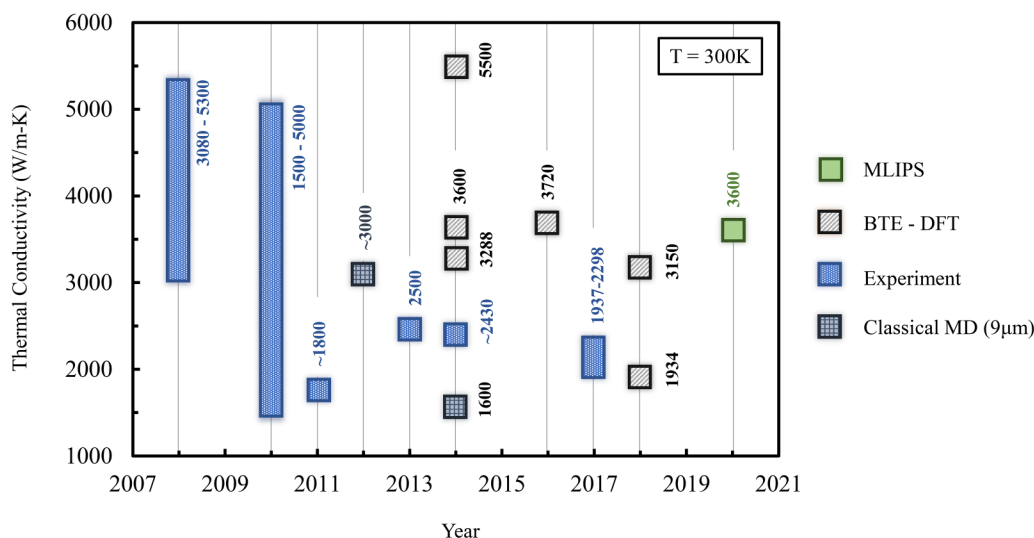


FIG. 2. Graphene thermal conductivity by MLIPs, experiment, BTE-DFT, and classical MD approaches at room temperature reported in recent years.

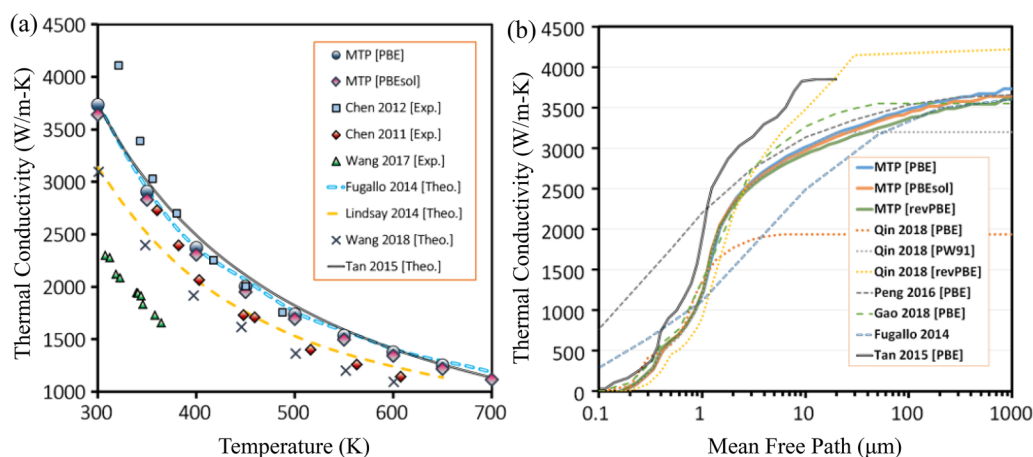


FIG. 3. (a) Graphene thermal conductivity vs temperature by the MTP-based solutions (MLIPs) compared with experimental (Exp.) works and theoretical (Theo.) results. (b) Accumulated thermal conductivity vs the mean free path of graphene at room temperature by MTP (MLIPs) and DFT approach by Qin *et al.*,⁸⁹ Fugallo *et al.*,^{87,88} Peng *et al.*,⁸⁶ Gao *et al.*,⁹⁶ and Tan *et al.*¹²⁰ with different exchange-correlation functions of Perdew–Burke–Ernzerhof (PBE), revised PBE (revPBE), PBE revised for solids (PBEsol), and Perdew–Wang 91 (PW91) [Reproduced with permission from Mortazavi *et al.*, Comput. Phys. Commun. **258**, 107583 (2021).⁵⁷ Copyright 2021 Elsevier].

substantially increase the computational costs and lead to complexity of the problem.⁵⁸ Even in the case of a highly symmetrical lattice of graphene, the MLIP-based method is highly efficient from the computational point of view.⁵⁷ Nevertheless, the computational efficiency of the MLIP-based solution will be even more obvious when moving from high- to low-symmetry structures and, in particular, for 2D lattices that require the consideration of interactions with distant neighbors.

B. Thermal conductivity of 3D materials

The proposed MLIPs' accuracy has been also investigated for different bulk materials. In this regard, the phononic properties of diamond, silicon, BAs, and InAs were studied using MLIPs and

BTE approaches. Figure 6⁵⁷ shows that the predicted phonon dispersion relations for the above-mentioned bulk materials by passively fitted MTPs are in close agreement with DFT-based results.

Figure 7 and Table III show the estimated phononic thermal conductivity of the bulk structures by the MLIPs, compared with experimental, full-DFT, and classical MD counterparts. The negligible computational cost of the force evaluations with MTP enables using relatively large supercells. The thermal conductivities obtained with MLIPs' approach are in good agreement with experimental, DFT-based theoretical, and classical MD results. The discrepancy occurred for the case of diamond, silicon, InAs, and BAs around 4%, 5%, 6%, and 12%, respectively, which can be related to computational details; for example, in the study by Protik *et al.*,¹²² they predicted a room temperature thermal conductivity of

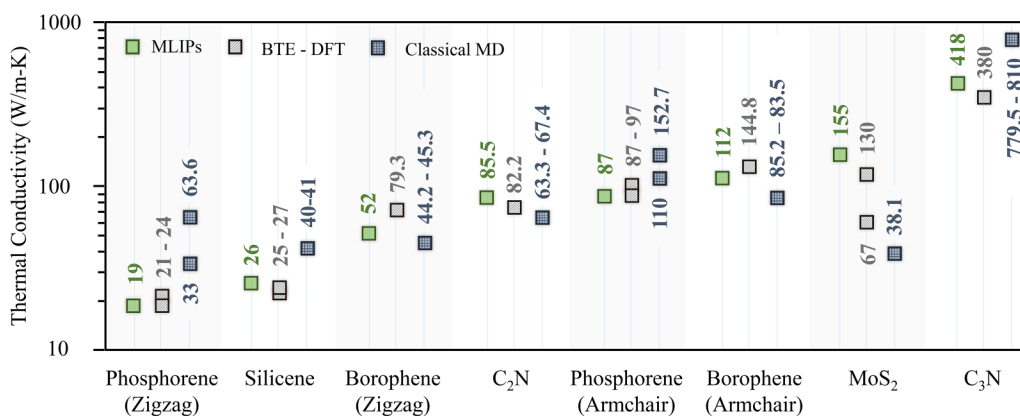


FIG. 4. Thermal conductivity of reported 2D nanomaterials by MLIPs, BTE–DFT, and classical MD approaches at room temperature.

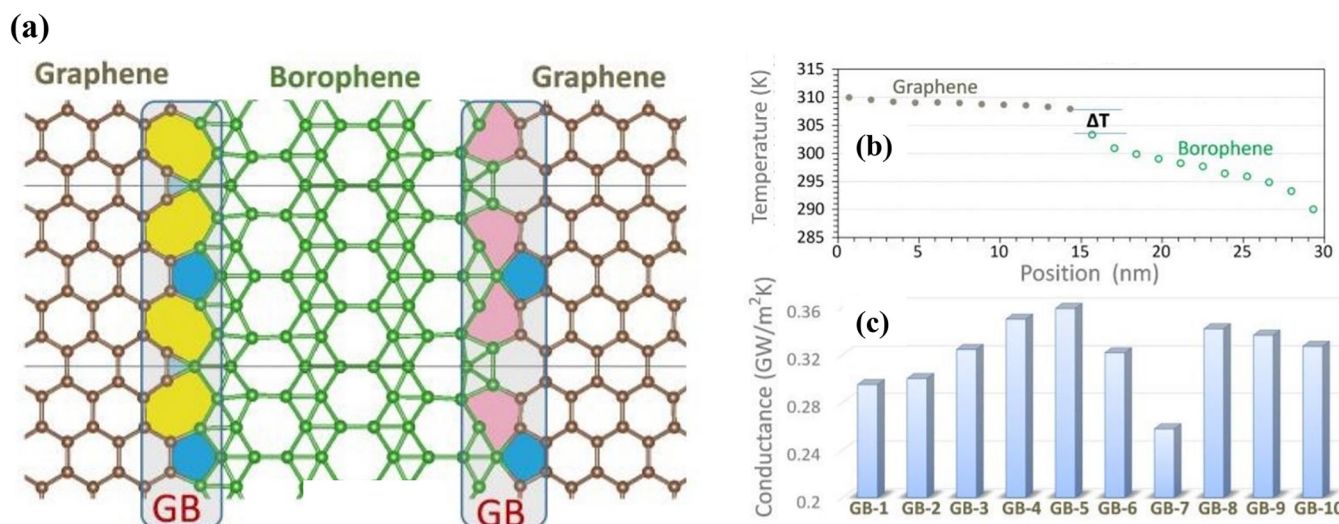


FIG. 5. (a) Atomic structure of constructed borophene/graphene grain boundaries (GB), (b) temperature profile, and (c) interfacial thermal conductance of grain boundaries [Reproduced with permission from Mortazavi *et al.*, *Mater. Horizons*, **7**, 2359–2367 (2020).⁴¹ Copyright 2020 Royal Society of Chemistry].

2300 W/m K for naturally occurring BAs, which is only by around 9% difference from the MTP-based predicted value.⁵⁷ Interestingly, Yang *et al.*¹¹⁶ demonstrated that machine learning-based potential can be a promising method for modeling the thermal conductivity of both crystalline and amorphous materials with strong disorders. The thermal conductivity of crystalline silicon and amorphous silicon at room temperature is reported to be 121 and 1.4 W/mK, respectively, using EMD, which is in good agreement with the results of experiments and first-principles calculations.

VI. FUTURE PERSPECTIVE

A. Mechanical properties

Besides the extensive application of MLIPs in studying phononic properties of nanomaterials, it can be a useful means to calculate other characteristics such as mechanical properties (e.g.,

elastic modulus, fractural point). It has been shown that the MLIPs can well predict the mechanical properties of C₂N,⁵⁴ graphene-like BC₆N using MD approaches,¹²⁵ and graphene/borophene heterostructures by first-principles modeling.¹²

In addition to their high potential in simulating and evaluating the properties of the pristine form of nanomaterials, MLIPs have been proven to be effective enough in predicting characteristics of defective nanomaterials. Recently, the mechanical properties of C₂N with the existence of point defects have been estimated by MLIPs.^{54,125}

B. MLIPs' challenges

The MLIP method is still associated with viable challenges that need to be addressed. Reference datasets based on the electronic structure calculations are the necessities of training MLIPs that make the process computationally difficult and

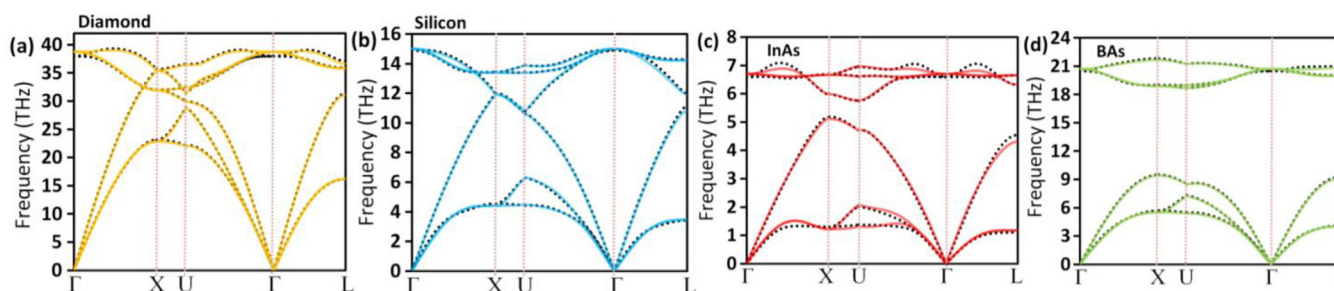


FIG. 6. Phonon dispersion relations of (a) bulk diamond, (b) silicon, (c) InAs, and (d) BAs estimated using the DFT (dotted lines) and MLIPs (continuous lines). [Reproduced with permission from Mortazavi *et al.*, *Comput. Phys. Commun.* **258**, 107583 (2021).⁵⁷ Copyright 2021 Elsevier].

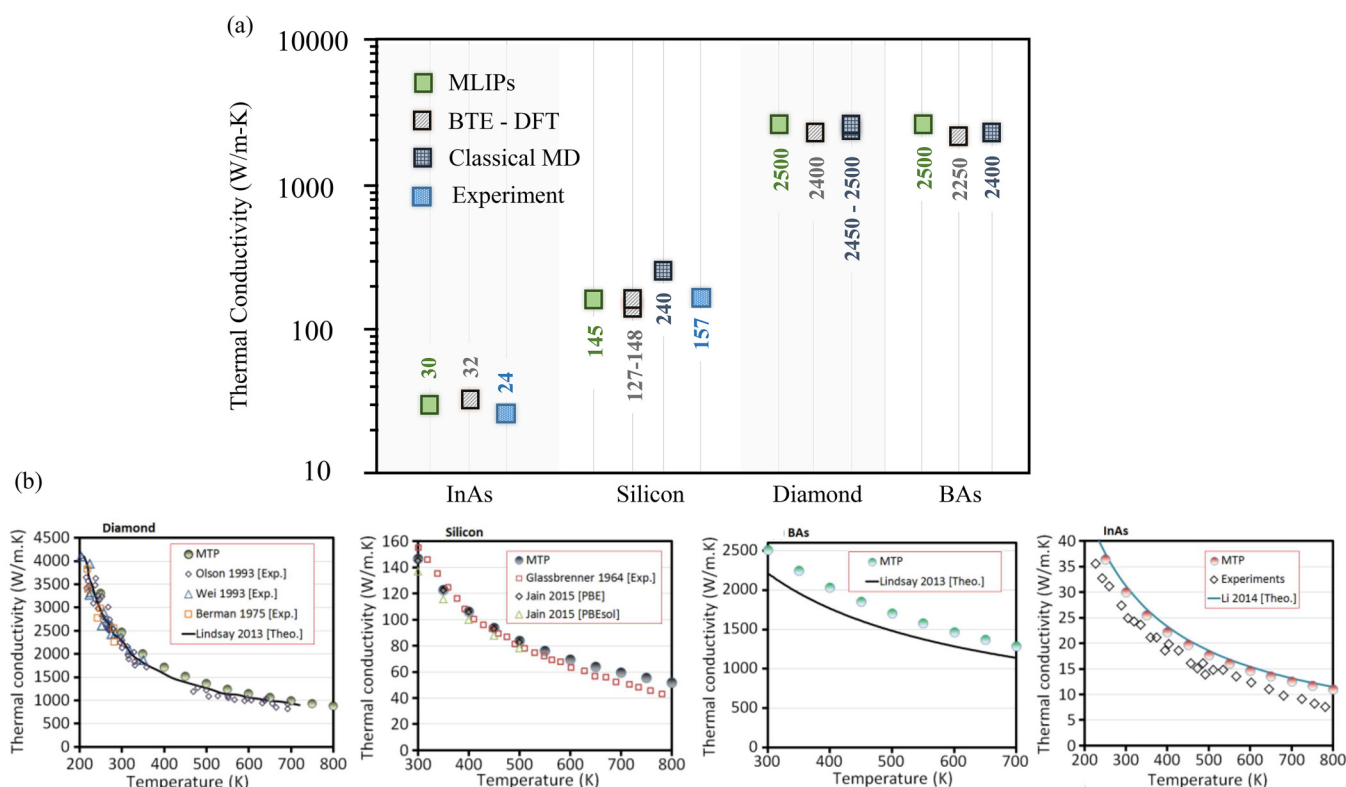


FIG. 7. Thermal conductivity of reported 3D materials by MLIPs, experiment, BTE-DFT and classical MD approaches (a) at room temperature, (b) at different temperatures by Olson *et al.*,¹²³ Wei *et al.*,¹¹¹ Berman *et al.*,¹¹⁰ Lindsay *et al.*,¹¹² and Chen *et al.*¹²⁴ for diamond, Glassbrenner and Slack¹¹³ and Jain and McGaughey¹¹⁴ for silicon, Lindsay *et al.*¹¹² for BaS, and Li *et al.*⁷⁴ for InAs [(b) Reproduced with permission from Mortazavi *et al.*, *Comput. Phys. Commun.* **258**, 107583 (2021).⁵⁷ Copyright 2021 Elsevier].

time-consuming. All available information about the potential energy surface (PES), like forces, has to be utilized regarding downsizing the reference datasets, which should therefore be one of the main targets. Minimizing the reference datasets is a novel method at the beginning of the way that depends on the intricacy of the PES, which needs to be mapped in all its details. Anyhow, since machine learning potentials are likely to collapse if they are not trained correctly, they demand rigorous trials and validations. All of the desired states and configurations have to be included in the training set; otherwise, MLIPs are not able to provide new insights. The complexity of configuration space rises exponentially with the chemical structure and leads to the increment of either the number of descriptors or the size of the reference dataset, which is another deficiency of this approach. Lots of MLIPs are therefore confined to not many chemical elements and other potential functions in materials science that suffer from the same limitation. A high chemical complication cannot be handled unless the configuration space is diminished, which is possible by each of using low-dimensional terms same as classical force fields or by limiting the local minima of the PES. MLIPs improve the accuracy of large-scale MD simulations to the level of quasi-*ab initio* ones. Consequently, this level of accuracy commonly demands the development or

expansion of the reference dataset, which entails a great deal of effort. It can be therefore concluded that in the training process of an MLIP, most of the time is specialized to the preparation of the dataset.⁴⁶ In summary, drawbacks of MLIPs, to the best of our knowledge, can be categorized as follows:

- **Extrapolation:** If areas of configurational space are missing in the dataset, MLIPs fail to extrapolate the PES for them.
- **Scattered datasets:** From a numerical point of view, training an MLIP becomes significantly intricate when the initial training dataset comprises a variety of configurations and a diversity of energies like in the case of bulk cells next to surfaces. So, lots of current research studies in this area are held to find and implement precise corrections to optimize MLIPs.
- **Long-range interactions:** Atomic interactions are near-sighted. To elaborate, the governing contributions to the potential energy surface here are from atoms that are close to each other. Although sometimes the cutoff radius is larger and long-range interactions like dispersion interactions should be considered.
- **Chemical complexity:** It is hard to simulate multi-component structures, such as ternary or higher chemical alloys and compounds, with MLIPs.

- *The accuracy of the underlying reference data:* To examine the exactness of MLIPs in the training process, they are typically checked with resulted forces and energies of the DFT calculations, while the reference datasets are usually considered to be accurate. Yet, in many cases, the DFT calculations are not possible or collapse, which consequently leads to failure of MLIP training, and it is fiercely rare for MLIP results to be more aligned with experimental results than with the DFT ones.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors have no conflicts to disclose.

DATA AVAILABILITY

Data sharing is not applicable to this article as no new data were created or analyzed in this study.

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